# Homework 8

EE 999 — Introduction to Control Systems

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*Due Wednesday, November 11, 2026*

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## Problem A — PI and PD Control

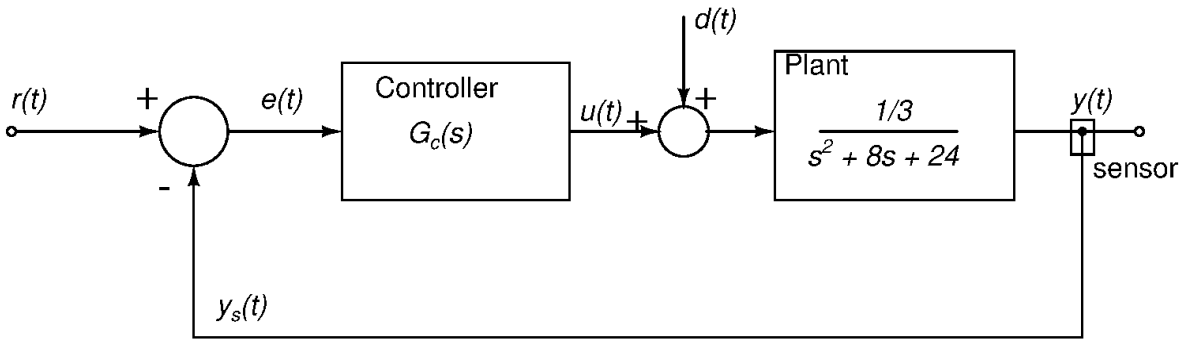



Figure 1: A system with PI-type control.

## Problem A.1 — PI Control

*Use* $G_c(s) = \dfrac{k_p s + k_i}{s}$ *with* $k_p = 10$, $k_i = 200$.

## Problem A.1.a

*Determine the transfer functions:*

**i)** $T_{ry}(s)$**, the input-output transfer function**

$$G_p(s) = \frac{1/3}{s^2 + 8s + 24} \tag{1}$$

$$G_c(s) = \frac{k_p s + k_i}{s} \tag{2}$$

$$Y(s) = \frac{G_p G_c}{1 + G_p G_c} R(s) \tag{3}$$

$$Y(s) = \frac{\dfrac{1/3}{s^2 + 8s + 24} \cdot \dfrac{10s + 200}{s}}{1 + \dfrac{1/3}{s^2 + 8s + 24} \cdot \dfrac{10s + 200}{s}} R(s) \tag{4}$$

$$Y(s) = \frac{\frac{1}{3}(10s + 200)}{s(s^2 + 8s + 24) + \frac{1}{3}(10s + 200)} R(s) \tag{5}$$

$$Y(s) = \frac{10s + 200}{3s(s^2 + 8s + 24) + 10s + 200} R(s) \tag{6}$$

$$Y(s) = \frac{10s + 200}{3s^3 + 24s^2 + 82s + 200} R(s) \tag{7}$$

$$T_{ry}(s) = \frac{Y(s)}{R(s)} = \frac{10s + 200}{3s^3 + 24s^2 + 82s + 200} \tag{8}$$

**ii)** $T_{re}(s)$**, the error transfer function**

$$E(s) = R(s) - Y(s) \tag{9}$$
$$E(s) = R(s) - T_{ry}(s)R(s) \tag{10}$$
$$E(s) = \left(1 - \frac{10s + 200}{3s^3 + 24s^2 + 82s + 200}\right) R(s) \tag{11}$$
$$E(s) = \left(\frac{3s^3 + 24s^2 + 82s + 200}{3s^3 + 24s^2 + 82s + 200} - \frac{10s + 200}{3s^3 + 24s^2 + 82s + 200}\right) R(s) \tag{12}$$
$$E(s) = \frac{3s^3 + 24s^2 + 72s}{3s^3 + 24s^2 + 82s + 200} R(s) \tag{13}$$
$$T_{re}(s) = \frac{E(s)}{R(s)} = \frac{3s^3 + 24s^2 + 72s}{3s^3 + 24s^2 + 82s + 200} \tag{14}$$

**iii) $T_{dy}(s)$, the disturbance transfer function**

$$Y(s) = G_p \left(D(s) + U(s)\right) \tag{15}$$
$$Y(s) = G_p \left(D(s) + G_c(s)E(s)\right) \tag{16}$$
$$Y(s) = G_p \left(D(s) - G_c(s)Y(s)\right) \tag{17}$$
$$Y(s) = G_p D(s) - G_p G_c(s) Y(s) \tag{18}$$
$$Y(s) + G_p G_c(s) Y(s) = G_p D(s) \tag{19}$$
$$Y(s) \left(1 + G_p G_c(s)\right) = G_p D(s) \tag{20}$$
$$Y(s) = \frac{G_p}{1 + G_p G_c(s)} D(s) \tag{21}$$
$$Y(s) = \frac{\dfrac{1/3}{s^2 + 8s + 24}}{1 + \dfrac{1/3}{s^2 + 8s + 24} \cdot \dfrac{10s + 200}{s}} D(s) \tag{22}$$
$$Y(s) = \frac{\frac{1}{3}s}{s(s^2 + 8s + 24) + \frac{1}{3}(10s + 200)} D(s) \tag{23}$$
$$Y(s) = \frac{s}{3s(s^2 + 8s + 24) + 10s + 200} D(s) \tag{24}$$
$$Y(s) = \frac{s}{3s^3 + 24s^2 + 72s + 10s + 200} D(s) \tag{25}$$
$$Y(s) = \frac{s}{3s^3 + 24s^2 + 82s + 200} D(s) \tag{26}$$
$$T_{dy}(s) = \frac{Y(s)}{D(s)} = \frac{s}{3s^3 + 24s^2 + 82s + 200} \tag{27}$$

## Problem A.1.b

*Draw a pole-zero diagram of $T_{ry}(s)$, label the dominant mode.*

**Code:** `hw_08_A1b.m`

```
if exist('OCTAVE_VERSION', 'builtin')
    pkg load control;
end

num = [10 200];
den = [3 24 82 200];
sys = tf(num, den);

pzmap(sys);

sys_poles = pole(sys)
sys_zeros = zero(sys)
```

**Output:** `output/hw_08_A1b.txt`

```
sys_poles =

  -5.2097 +      0i
  -1.3952 + 3.2940i
  -1.3952 - 3.2940i

sys_zeros = -20
```

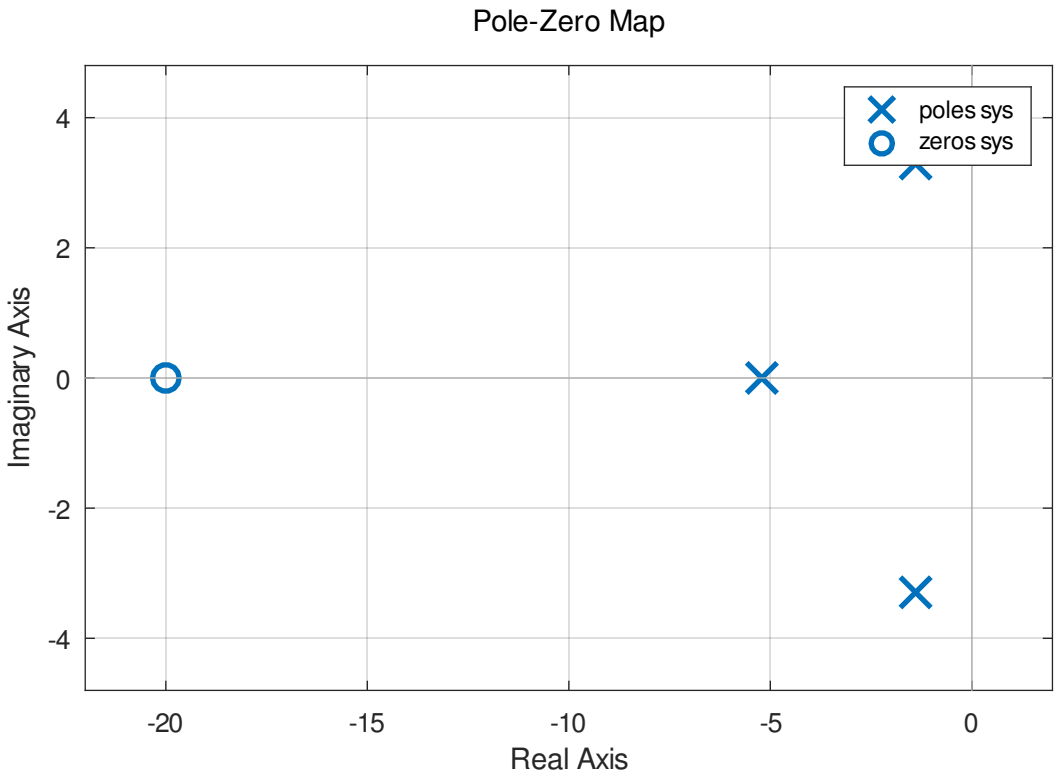



Figure 2: Pole-zero diagram of $T_{ry}(s)$.

The dominant poles are at $s = -1.3952 \pm 3.2940j$.

## Problem A.2

*Based on the dominant mode pole locations, determine*

$$p_{1,2} = -\alpha \pm j\omega \tag{28}$$
$$= -1.3952 \pm 3.2940j \tag{29}$$

$$\omega_n = \sqrt{\alpha^2 + \omega^2} \tag{30}$$
$$= \sqrt{1.3952^2 + 3.2940^2} \tag{31}$$
$$= 3.5773 \text{ rad/s} \tag{32}$$

**i) $t_p$, the peak time**

$$t_p = \frac{\pi}{\omega} \tag{33}$$
$$= \frac{\pi}{3.2940} \tag{34}$$
$$= 0.9537 \text{ s} \tag{35}$$

**ii) $\rho$, the damping**

$$\rho = \frac{\alpha}{\omega_n} \tag{36}$$
$$= \frac{1.3952}{3.5773} \tag{37}$$
$$= 0.3900 \tag{38}$$

**iii) $M_p$, the percent overshoot**

$$M_p = 100\% \times e^{-\pi\alpha/\omega} \tag{39}$$
$$= 100\% \times e^{-\pi(1.3952)/3.2940} \tag{40}$$
$$= 26.43\% \tag{41}$$

**iv) $t_r$, the rise time**

$$t_r \approx \frac{1.8}{\omega_n} \tag{42}$$
$$\approx \frac{1.8}{3.5773} \tag{43}$$
$$\approx 0.5032 \text{ s} \tag{44}$$

**v) $t_s$, the settling time**

$$t_s \approx \frac{4}{\alpha} \tag{45}$$
$$\approx \frac{4}{1.3952} \tag{46}$$
$$\approx 2.8670 \text{ s} \tag{47}$$

## Problem A.3

*Analytically determine $e_{ss}$, the steady state error, by determining the system type and $K_{Bode}$.*

$$G_L(s) = G_c(s)\, G_p(s)\, H_y(s) \tag{48}$$
$$= \frac{10s + 200}{s} \cdot \frac{1/3}{s^2 + 8s + 24} \cdot 1 \tag{49}$$
$$= \frac{\frac{1}{3}(10s + 200)}{s(s^2 + 8s + 24)} \tag{50}$$

$G_L(s)$ has one pole at the origin, so the system is Type I. A Type I system has no steady-state error for a step input, $e_{ss} = 0$.

## Problem A.4

*Setup $T_{ry}(s)$ in Matlab, run and plot the (true) step response. Determine these values for the complete system*

$$T_{ry}(s) = \frac{10s + 200}{3s^3 + 24s^2 + 82s + 200} \tag{8}$$

**Code:** hw_08_A4.m

```
if exist('OCTAVE_VERSION', 'builtin')
    pkg load control;
end

% T_ry(s) = (10s + 200) / (3s^3 + 24s^2 + 82s + 200)
num = [10 200];
den = [3 24 82 200];
sys = tf(num, den);

info = stepinfo(sys)

damp(sys)

step(sys);
grid on;
title('Step response of T_{ry}(s)');
```

**Output:** output/hw_08_A4.txt

```
info =

  scalar structure containing the fields:

    RaiseTime = [](0x0)
    RiseTime = 0.4936
    TransientTime = 2.4754
    SettlingTime = 2.4754
    SettlingMin = 0.9000
    SettlingMax = 1.2038
    Overshoot = 20.376
    Undershoot = 0
    Peak = 1.2038
    PeakTime = 1.1429
    StaticGain = 1
    InitialJump = 0

   Pole                  Damping     Frequency       Time Constant
                                     (rad/seconds)   (seconds)

   -1.40e+00+3.29e+00i   3.90e-01    3.58e+00        7.17e-01
   -1.40e+00-3.29e+00i   3.90e-01    3.58e+00        7.17e-01
   -5.21e+00+0.00e+00i   1.00e+00    5.21e+00        1.92e-01
```

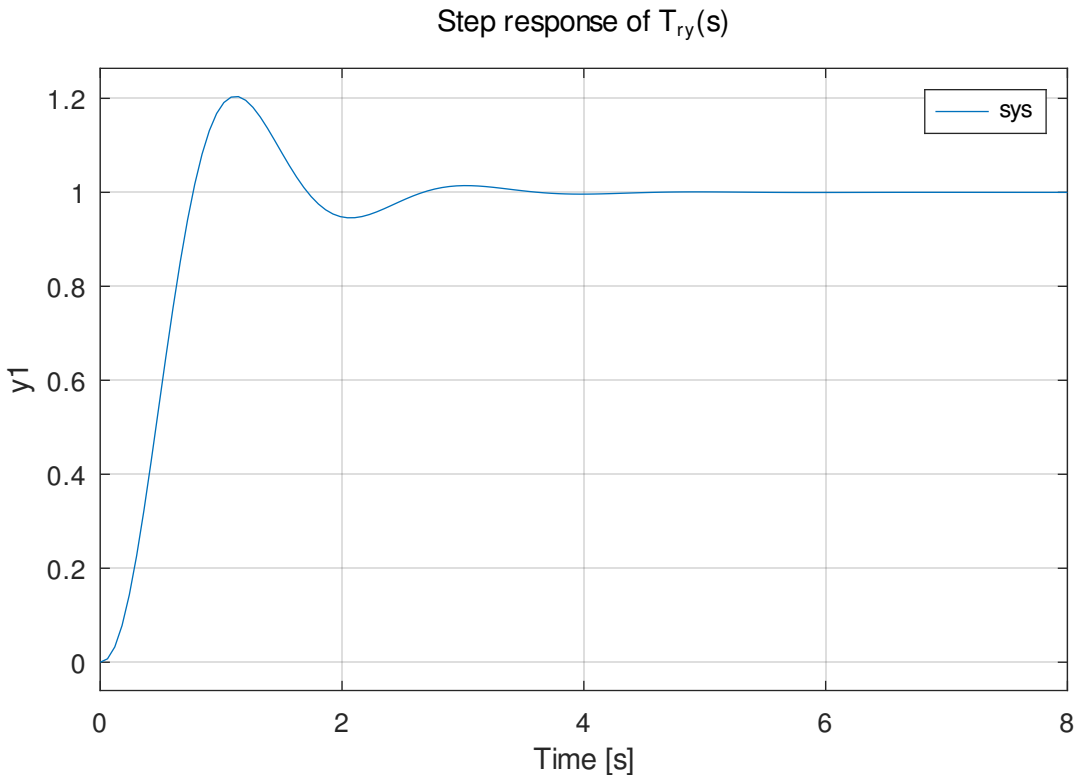



Figure 3: Step response of $T_{ry}(s)$.

**i) $t_p$, the peak time**

$$t_p = 1.1429 \text{ s} \tag{51}$$

**ii) $\rho$, the damping**

$$\rho = 0.3900 \tag{52}$$

**iii) $M_p$, the percent overshoot**

$$M_p = 20.38\% \tag{53}$$

**iv) $t_r$, the rise time**

$$t_r = 0.4936 \text{ s} \tag{54}$$

**v) $t_s$, the settling time**

$$t_s = 2.4754 \text{ s} \tag{55}$$

**vi) $e_{ss}$, the steady state error**

$$e_{ss} = 0 \tag{56}$$

## Problem A.5

*(Essay question) What is the value of the methods of steps A.1.b–A.2, relative to the method of step A.4?*

The value of the methods in A.1.b–A.2 is that they provide reasonable estimates just by evaluating the dominant pole pair without the rigor of actually having to plot or otherwise analyze the entire system response. The tradeoff is that the numbers are not accurate.

## Problem A.6

*Based on parts A.1.b–A.4, which measures based on the dominant modes came out most accurate? Which measures based on the dominant modes (part 3) came out least accurate?*

| Measure | Estimate | Actual | Error |
|---|---|---|---|
| $t_p$, peak time | 0.9537 s | 1.1429 s | −16.6% |
| $\rho$, damping | 0.3900 | 0.3900 | 0.0% |
| $M_p$, percent overshoot | 26.43% | 20.38% | +29.7% |
| $t_r$, rise time | 0.5032 s | 0.4936 s | +1.9% |
| $t_s$, settling time | 2.8670 s | 2.4754 s | +15.8% |
| $e_{ss}$, steady-state error | 0 | 0 | – |

Table 1: Dominant-mode estimates (A.2, A.3) compared with the complete-system values (A.4). Error is (estimate − actual)/actual.

The damping and steady-state error were the most accurate, and the percent overshoot was the least accurate.

## Problem B — PD Control

*Use $G_c(s) = k_d s + k_p$ with $k_p = 300$, and $k_d$ to be determined.*

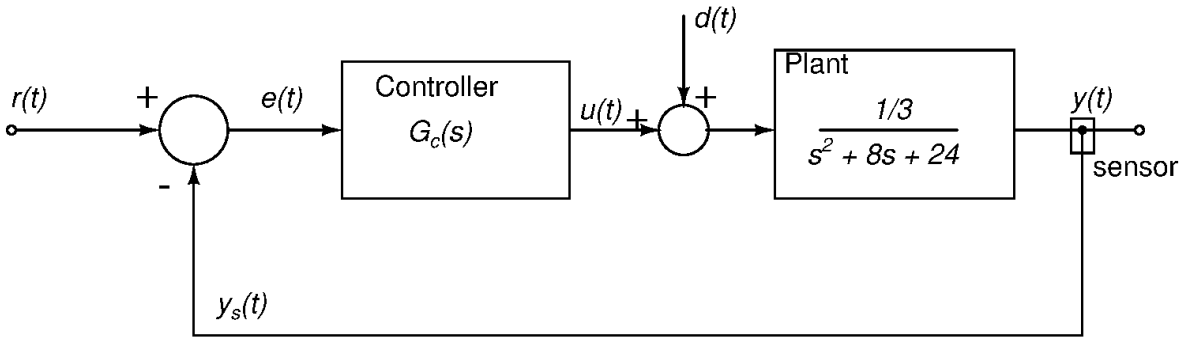



Figure 1: The system from Problem A, repeated, with PD control $G_c(s) = k_d s + k_p$ in place of PI.

## Problem B.1 — Experiment with $k_d$

### Problem B.1.a

*Derive $T_{ry}(s)$, the closed loop transfer function with PD control, and with $k_p$ and $k_d$ as variables. Set $k_p = 300$ and determine the pole locations for several values of $k_d$ on the range from 0.5 to 10.*

$$G_c(s) = k_d s + k_p \tag{57}$$

$$G_p(s) = \frac{1/3}{s^2 + 8s + 24} \tag{58}$$

$$Y(s) = \frac{G_p G_c}{1 + G_p G_c} R(s) \tag{59}$$

$$Y(s) = \frac{\dfrac{1/3}{s^2 + 8s + 24} \cdot (k_d s + k_p)}{1 + \dfrac{1/3}{s^2 + 8s + 24} \cdot (k_d s + k_p)} R(s) \tag{60}$$

$$Y(s) = \frac{\frac{1}{3}(k_d s + k_p)}{s^2 + 8s + 24 + \frac{1}{3}(k_d s + k_p)} R(s) \tag{61}$$

$$Y(s) = \frac{k_d s + k_p}{3(s^2 + 8s + 24) + k_d s + k_p} R(s) \tag{62}$$

$$Y(s) = \frac{k_d s + k_p}{3s^2 + 24s + 72 + k_d s + k_p} R(s) \tag{63}$$

$$Y(s) = \frac{k_d s + k_p}{3s^2 + (24 + k_d)s + (72 + k_p)} R(s) \tag{64}$$

$$T_{ry}(s) = \frac{Y(s)}{R(s)} = \frac{k_d s + k_p}{3s^2 + (24 + k_d)s + (72 + k_p)} \tag{65}$$

**Code:** `hw_08_B1a.m`

```
if exist('OCTAVE_VERSION', 'builtin')
    pkg load control;
end

k_p = 300;

for k_d = [0.5 2 5 8 10]
    k_d

    num = [k_d k_p];
    den = [3 24+k_d 72+k_p];

    sys = tf(num, den);
    k_d_poles = pole(sys)
end
```

**Output:** `output/hw_08_B1a.txt`

```
k_d = 0.5000
k_d_poles =

   -4.083 + 10.360i
   -4.083 - 10.360i

k_d = 2
k_d_poles =

   -4.333 + 10.258i
   -4.333 - 10.258i

k_d = 5
k_d_poles =

   -4.833 + 10.032i
   -4.833 - 10.032i

k_d = 8
k_d_poles =

  -5.3333 + 9.7753i
  -5.3333 - 9.7753i

k_d = 10
k_d_poles =

  -5.6667 + 9.5859i
  -5.6667 - 9.5859i
```

## Problem B.1.b

*Hand sketch a plot of the closed-loop pole locations as a function of $k_d$.*

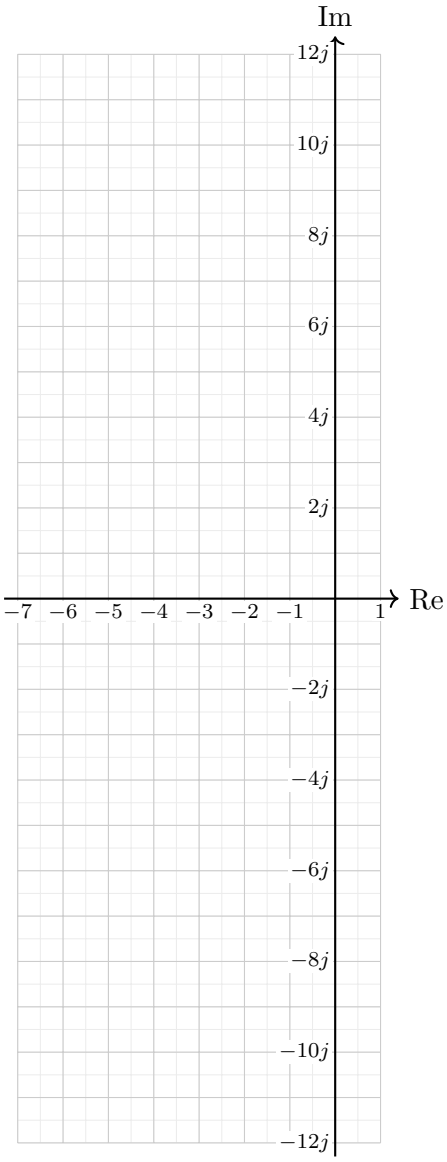



Figure 4: Closed-loop poles of $T_{ry}(s)$ as $k_d$ varies from 0.5 to 10, with $k_p = 300$.

## Problem B.1.c

*Select a value of $k_d$ to give $\rho = 0.4$, be rather accurate, e.g., $0.39 < \rho < 0.41$.*

**Code:** `hw_08_B1c.m`

```
if exist('OCTAVE_VERSION', 'builtin')
    pkg load control;
end

k_p = 300;
rho_err = .1;

for k_d = 2:0.1:3
```

```
    num = [k_d k_p];
    den = [3 24+k_d 72+k_p];

    sys = tf(num, den);
    [w_n, rho, p] = damp(sys);

    rho_test = abs(0.4-rho(1));

    if rho_test < rho_err
        rho_err = rho_test;
        k_d_best = k_d;
        rho_best = rho(1);
        p_best = p;
    end
end

k_d_best
rho_best
p_best
```

**Output:** `output/hw_08_B1c.txt`

```
k_d_best = 2.7000
rho_best = 0.3996
p_best =

   -4.450 + 10.208i
   -4.450 - 10.208i
```

Using the value of $k_d$ chosen to give $\rho = 0.4$:

## Problem B.2

*What is the dominant mode of this system? (Same as asking "where are the dominant pole(s)")*

With $k_d = 2.7$:

$$p_{1,2} = -\alpha \pm j\omega \tag{66}$$
$$= -4.4500 \pm 10.2077j \tag{67}$$

## Problem B.3

*Analyze your system (analytically, based on the dominant mode) to determine:*

**i) $t_p$, the peak time**

*write the answer here*

**ii) $\rho$, the damping**

*write the answer here*

**iii) $M_p$, the percent overshoot**

*write the answer here*

**iv) $t_r$, the rise time**

*write the answer here*

**v) $t_s$, the settling time**

*write the answer here*

## Problem B.4

*Analytically determine $e_{ss}$, the steady state error, by determining the system type and $K_{Bode}$.*

*write the answer here*

## Problem B.5

*Analyze the step response of your system (step response, full system, you can use the* `step` *and* `stepinfo` *commands) to determine:*

**i) $t_p$, the peak time**

*write the answer here*

**ii) $\rho$, the damping**

*write the answer here*

**iii) $M_p$, the percent overshoot**

*write the answer here*

**iv) $t_r$, the rise time**

*write the answer here*

**v) $t_s$, the settling time**

*write the answer here*

**vi) $e_{ss}$, the steady state error (not indicated by `stepinfo()`)**

*write the answer here*

## Problem B.6

*Which of the values based on dominant mode are closest to their analysis values? Which are farthest?*

*write the answer here*

## **Problem C** — Compare the PI and PD controllers

*Consider these parameters; the desirable characteristic is indicated.*

| Controller performance measure | Desirable characteristic |
|---|---|
| i) $t_p$, the peak time | Fast |
| ii) $\rho$, the damping | Near one |
| iii) $M_p$, the percent overshoot | Near zero |
| iv) $t_r$, the rise time | Fast |
| v) $t_s$, the settling time | Fast |
| vi) $e_{ss}$, the steady-state error | Small |

## **Problem C.1**

*What characteristic is most enhanced by the PI control system?*

*write the answer here*

## **Problem C.2**

*What characteristic is most enhanced by the PD control system (tuned for $\rho = 0.4$)?*

*write the answer here*